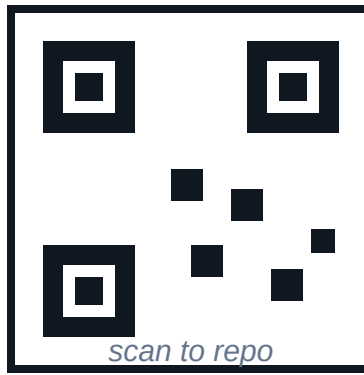




A Unified Control Plane for Machine-Learning Security Services

An Authenticated Gateway Fronting Heterogeneous ML Security Products

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Security Engineering • AI / Agent Security • Cloud Security

Apache-2.0 • Python 3.12 • HEAD 7ac348c • verified 2026-09-26

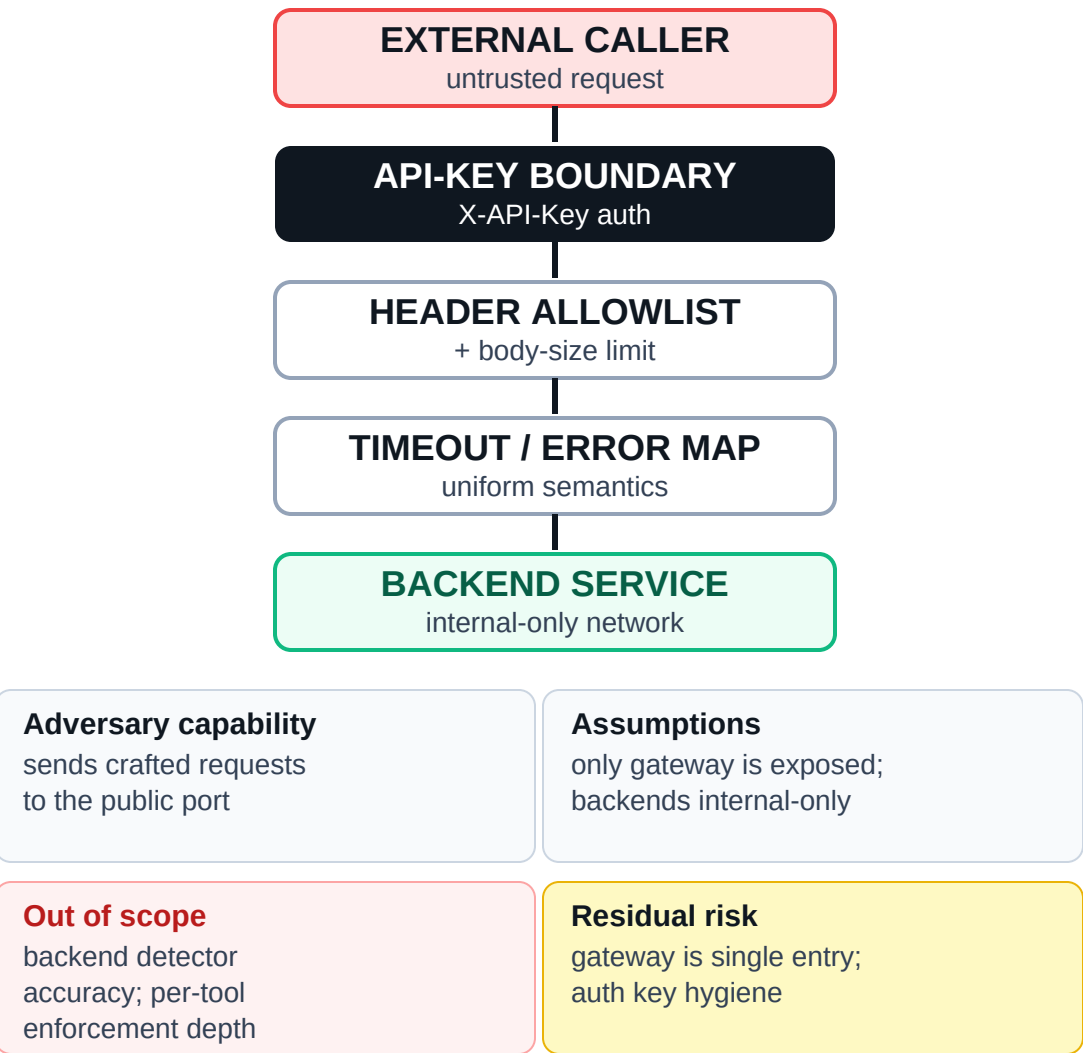
RESEARCH QUESTION

Can heterogeneous ML security services be fronted by one authenticated control plane with consistent auth, limits, and error handling?

01 • THE FAILURE MODE

Each ML security tool ships its own HTTP service, auth model, and error semantics. Wiring them into a product means every caller re-implements auth, header handling, body limits, and timeouts — inconsistently. A single gateway normalizes the entry boundary for the three real backend services.

02 • THREAT MODEL

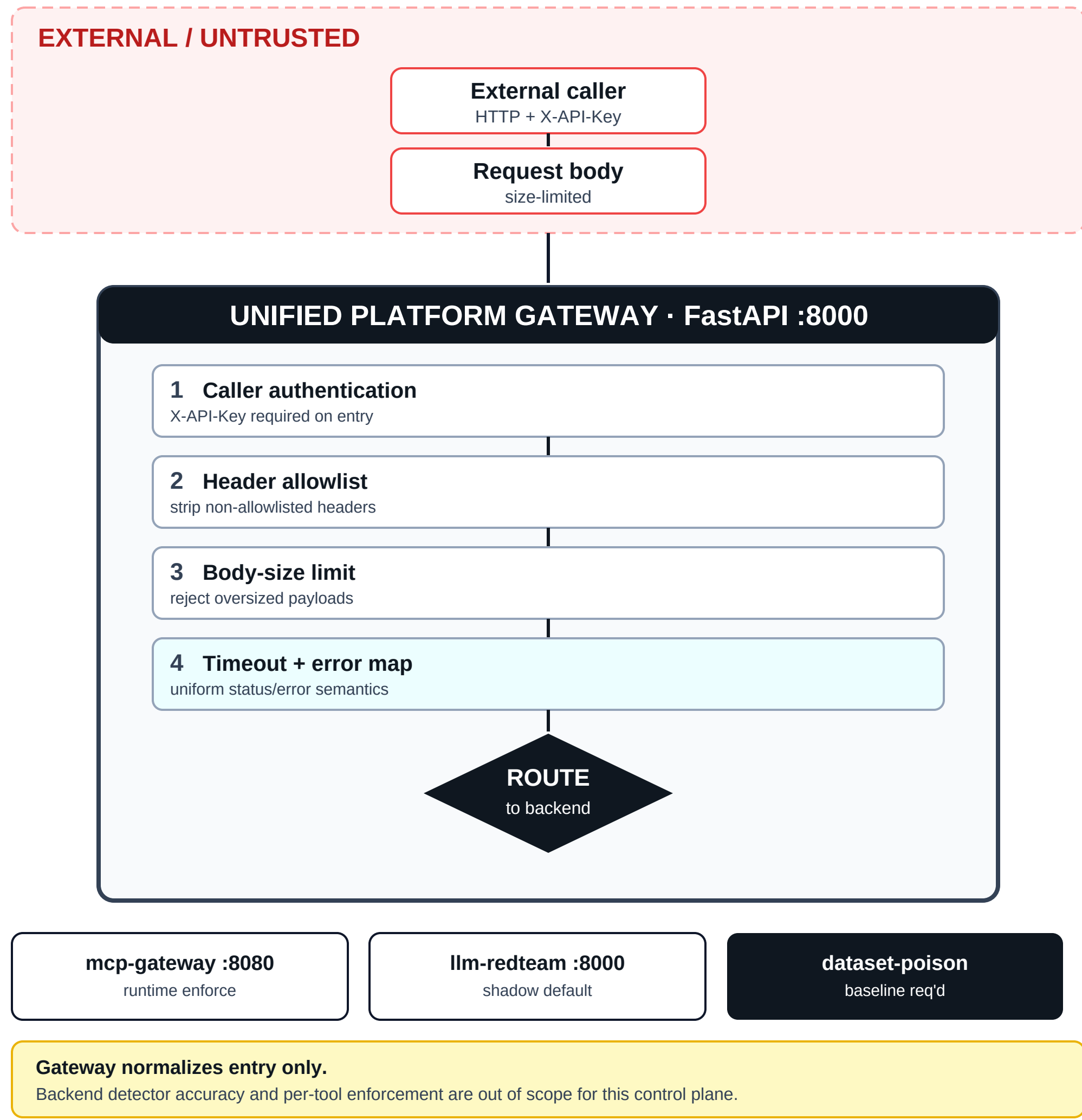


03 • OBJECTIVE

Determine whether one FastAPI gateway can provide consistent auth, allowlisting, limits, and error mapping across three ML security backends.

- 01 – X-API-Key caller auth
- 02 – Header allowlist + body-size limit
- 03 – Timeout + uniform error mapping
- 04 – Internal-only backend network

HERO • SYNCHRONOUS CONTROL PLANE



RESULTS • CURRENT LOCAL VALIDATION

59

TEST FUNCTIONS

3

REAL BACKENDS

59 test_ functions across the platform test suite.
Three integrated HTTP products behind one gateway.

Verified surface

Public host ports	1
Auth mechanism	API key
Compose files	2

ENTRY-BOUNDARY CONTROLS

Caller auth	100%
Header allowlist	100%
Body-size limit	100%
Error mapping	100%

Controls implemented at the gateway entry. Backend enforcement depth is per-tool, not sho

ARCHITECTURE PROVENANCE

ARCHITECTURE.md documents a synchronous control plane + separate batch/admission jobs. Production Docker network is internal-only; gateway alone exposed.

NOT ESTABLISHED BY THIS REPO

Backend detector accuracy. Distributed orchestration. Production reliability, uptime, or throughput SLOs.

SECURITY BOUNDARY

PROTECTS / DETECTS

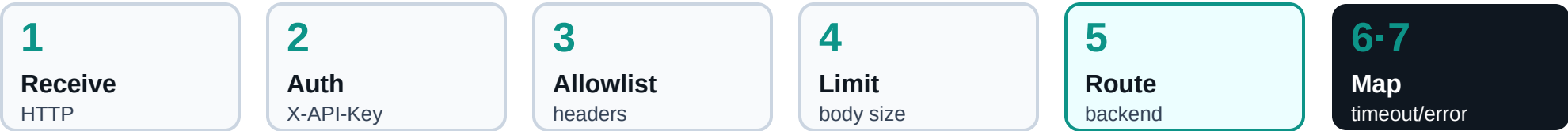
- authenticated entry
- header allowlist
- body-size limits
- uniform errors
- backend isolation

DOES NOT ESTABLISH

- backend accuracy
- async orchestration
- distributed scale
- production SLA
- per-tool enforce

Figure 1. Synchronous control plane: one authenticated FastAPI gateway fronts three real backends on an intern

METHOD • REQUEST HANDLING



ONE AUTHENTICATED ENTRY

CONTRIBUTION

A synchronous authenticated control plane that fronts three real ML security services (MCP gateway, LLM red-team, dataset-poison) with consistent auth, header allowlisting, body-size limits, and uniform error mapping on an internal-only Docker network.

LIMITATIONS

- 1. Gateway normalizes entry, not backend accuracy.
- 2. Synchronous control plane; not an async orchestrator.
- 3. Single public entry point (blast-radius consideration).
- 4. Local tests ≠ production reliability.
- 5. Per-tool enforcement depth varies (documented).

FUTURE WORK

- Async / queued orchestration mode.
- Per-backend rate limiting.
- Distributed multi-replica state strategy.
- End-to-end integration benchmark.
- mTLS between gateway and backends.

CONCLUSION

The platform provides one authenticated FastAPI gateway that normalizes auth, allowlisting, limits, and errors across three real ML security backends. It integrates services; it does not orchestrate them asynchronously or assert backend accuracy.

REPRODUCE

docker compose up -d
pytest tests/